MIL-I-81387A(AS)
2 April 1971
SUPERSEDING
MIL-I-81387(AS)
18 July 1966

MILITARY SPECIFICATION

INDICATORS, LIQUID OXYGEN QUANTITY

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - This specification covers liquid oxygen quantity indicators, having capacities of 5, 10, 20 or 25 liters, for aircraft liquid oxygen systems.

1.2 Classification - Indicators shall be of the following types (see 6.2):

Type I - 5 Liter	GMU-60/A
Type II - 10 Liter	GMU-61/A
Type III - 20 Liter	GMU-62/A
Type IV - 25 Liter	GMU-63/A

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications form a part of this specification. Unless otherwise specified, the issue in effect on date of invitation for bids shall apply.

SPECIFICATIONS

Federal

QQ-P-416 Plating, Cadmium (Electrodeposited)

Military

MIL-P-116 Preservation, Methods of

FSC 1660

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SPECIFICATIONS

Military (Cont'd)

MIL-T-152	Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-20708	Synchros, 60 and 400 Cycle, General Specification
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-I-81388	Indicator Repeaters, Liquid Oxygen Quantity

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

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STANDARDS

Military (Cont'd)

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MS 33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS 33585	Pointer, Dial, Standard Design of Aircraft Instrument
MS 33586	Metals, Definition of Dissimilar
MS 33639	Cases, Instrument, Clamp-mounted, Aircraft

(When requesting any of the applicable documents, refer to both title and number. All requests should be made via the cognizant Government quality control representative. Copies of this specification and other unclassified specifications and drawings required by contractors in connection with specific procurement functions should be obtained upon application to the Commanding Officer, Naval Supply Depot (Code 1051), 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. All other documents should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification - The indicator furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, and 6.4). In addition, the retention of the qualification for the indicator on the applicable qualified products list shall be dependent on periodic verification of continued compliance with the requirements of this specification (see 4.3 and 4.3.2).

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3.2 First article - The indicator furnished under this specification shall be a product which has been inspected and passed the first article inspection specified herein.

3.3 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials - Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts - All metal parts shall be of a corrosion resistance material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.2 Fungus proof materials - Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent. Electrical components shall be treated in accordance with MIL-T-152.

3.4.3 Protective treatment - When materials are used in the construction of the indicators that are subject to deterioration when exposed to environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. Protective coating which might crack, chip, or scale during normal service life or under extremes of environmental conditions shall not be used.

3.5 Standard parts - MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the manufacturer's drawings by their part numbers. Parts which are not covered by specifications or military standards, or which are not specifically described herein or on the drawings, shall be completely described on manufacturer's drawings.

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3.6 Design and construction - The indicator shall be designed to operate in conjunction with one or more liquid oxygen converters, having a capacitance type sensing element immersed in liquid oxygen, and shall be capable of driving a repeater indicator. The gauging system shall use a three-wire null-balancing capacitance bridge circuit with the sensing element of the converter connected into the sensing leg of the bridge. Solid state electronic circuits shall be used for amplification and power supply and shall be a part of the indicator. The electrical circuits shall contain a device to limit the current in the leads running to the sensing element to a maximum of 0.2 ampere during normal operation or in case of a short circuit. A servo motor shall drive the indicator pointer, a rebalancing potentiometer and a synchro transmitter that shall be used for driving a repeater indicator.

3.6.1 Low liquid level warning circuit - The indicator shall provide for a low liquid level warning circuit to operate a remote warning system at each station where required in the aircraft. The low level warning lamp shall be energized from a 5 volt AC external power source.

3.6.2 Press-to-test - The indicator shall incorporate a press-to-test circuit that shall drive the pointer down to zero position or below when the test switch is actuated. When the test switch is released, the pointer shall return to normal.

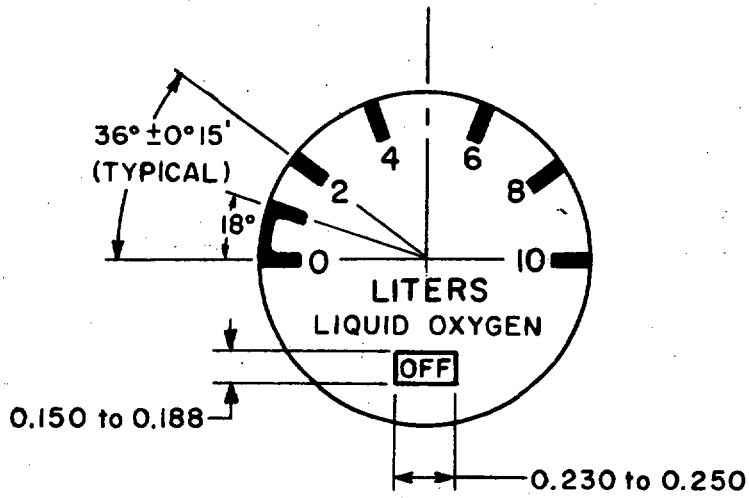
3.6.3 Power failure - The indicator shall provide for power failure indication. A power failure flag, in accordance with Figures 1 and 2, shall appear in the dial cut out when the input power to the indicator is interrupted. The power failure flag shown in Figures 1 and 2 is the preferred location; however, the dial cut out and the dial letters may be relocated provide they are not obscured by the pointer at any pointer position.

3.6.4 Indicator transmitting synchro - The indicator shall incorporate a synchro transmitter, in accordance with MIL-S-20708, as a means for driving a repeater indicator.

3.6.5 Empty and full calibration adjustments - The indicator shall be designed to operate without field adjustments; however, factory adjustment provisions shall be provided. The empty and full calibration adjustments shall be contained within the case; however, means for making empty and full adjustments shall be provided through the case.

3.6.5.1 Calibration adjustment seal - A seal shall be affixed to the "empty" and "full" calibration adjustments after such adjustments have been made.

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10 LITER DIAL



POINTER

IN ACCORDANCE WITH MS33585-8

DIAL DETAILS (ALL INDICATORS)	HEIGHT	PAINTING
NUMERALS	1/8"	MATTE WHITE FED-STD-595 COLOR NO.37875
GRADUATIONS	3/16" LONG BY .047" WIDE	
"LITERS"	1/8"	
"LIQUID OXYGEN"	3/32"	
"OFF"	3/32"	
POINTER SURFACE		
DIAL BACKGROUND		OPTICAL FLAT BLACK FED-STD-595 COLOR NO.37038
OFF FLAG BACKGROUND		RED FED-STD-595 COLOR NO.31136

LETTERS AND NUMERALS TO BE IN ACCORDANCE WITH MS33558

FIGURE 1. INDICATOR DIAL AND POINTER

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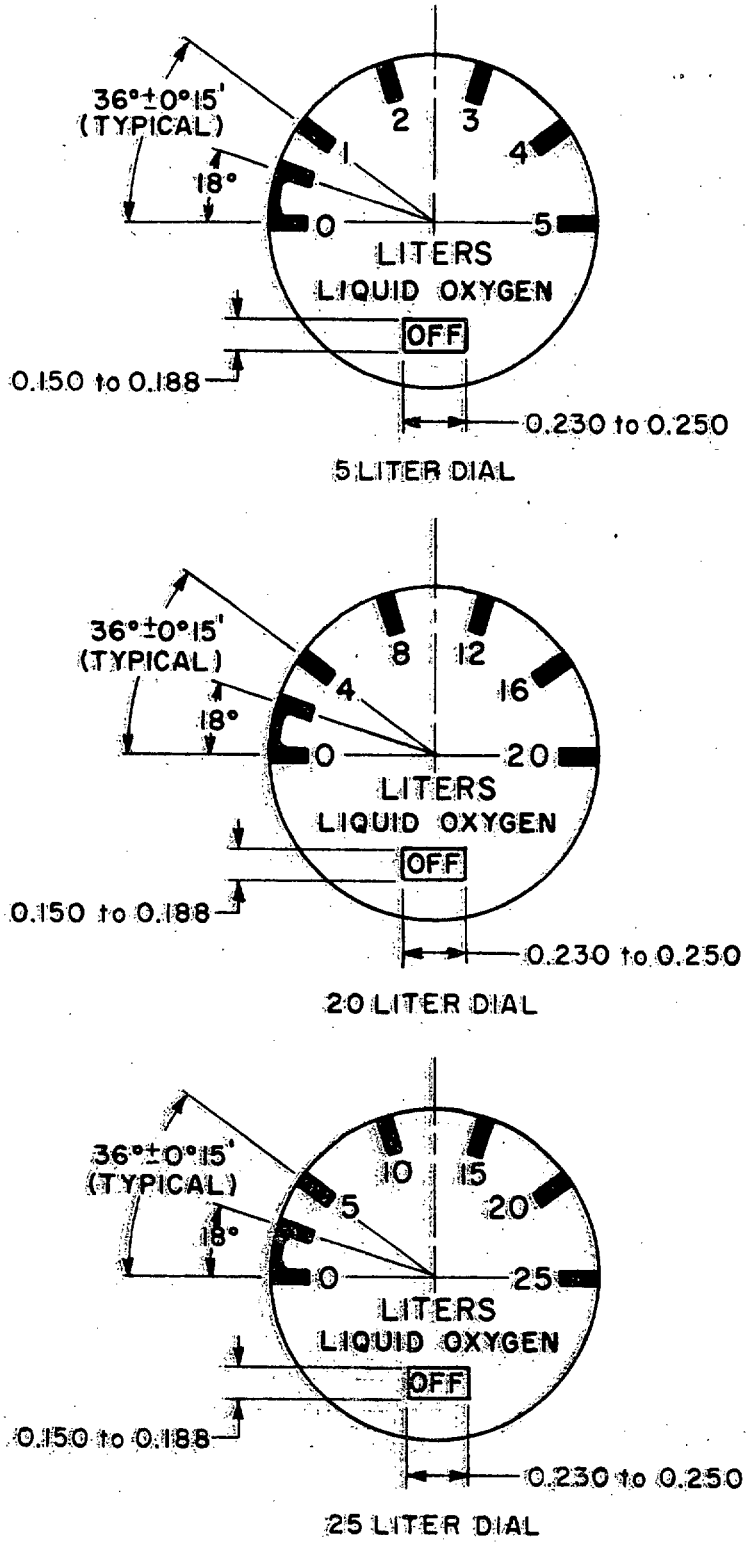


FIGURE 2. INDICATOR DIALS

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3.6.6 Case - The indicator case shall be in accordance with MS 33639 and Figure 3. The case shall be made of drawn metal suitable for hermetic sealing and of light construction.

3.6.7 Electrical receptacle - The electrical receptacle shall be wired internally in accordance with Figure 4 which is based upon clockwise rotation of the synchro rotor shaft for increasing quantities. If gears are used and pointer rotation is opposite to that of synchro shaft rotation, synchro transmitter stator connections, S_1 and S_3, shall be reversed at the receptacle to cancel the effects of the gear reversal.

3.6.8 Electronic parts - Solid state electronic circuit shall be used in the indicator. Electronic parts shall conform to MIL-E-5400.

3.6.9 Sealing - The indicator shall be hermetically sealed. The indicator shall be filled with an inert gas composed of a least 10 percent helium and 90 percent nitrogen. The gases shall be of at least 98 percent purity, free of dust particles, and containing not more than 0.006 milligrams of water vapor per liter (dew point minus 65°C) at the filling pressure. The absolute pressure of the gas in the case shall be slightly above atmospheric pressure.

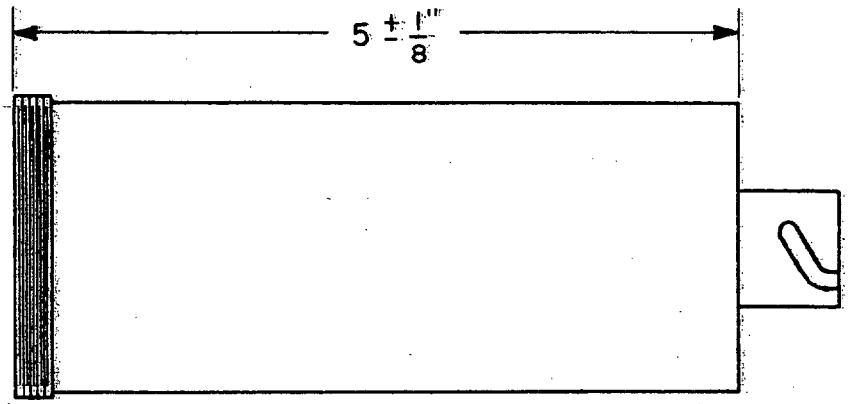
3.6.10 Dial - The indicator dial shall be fastened securely to the case or the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. If screws are used for attaching the dial, they shall be so located that they will not interfere with any dial marking. The pointer movement shall not be limited by a mechanical stop on the dial.

3.6.10.1 Dial markings - Dial markings shall conform to Figures 1 and 2. The style and proportions of the numerals and letters on the dial shall conform to MS 33558. Numerals shall distinctly indicate the graduation to which they apply.

3.6.10.2 Electrical zero reference mark - The electrical zero position of the synchro transmitter shall coincide with the zero quantity indication of the pointer.

3.6.11 Cover glass - The cover glass of the indicator shall be suitable for hermetic sealing and shall be flat, clear and free from flaws which might interfere with normal reading of the indicator. The distance between the front surface of the bezel and the outside surface of the glass shall not exceed 0.062 inch. The distance between the inside surface of the glass and the dial shall not exceed 0.156 inch.

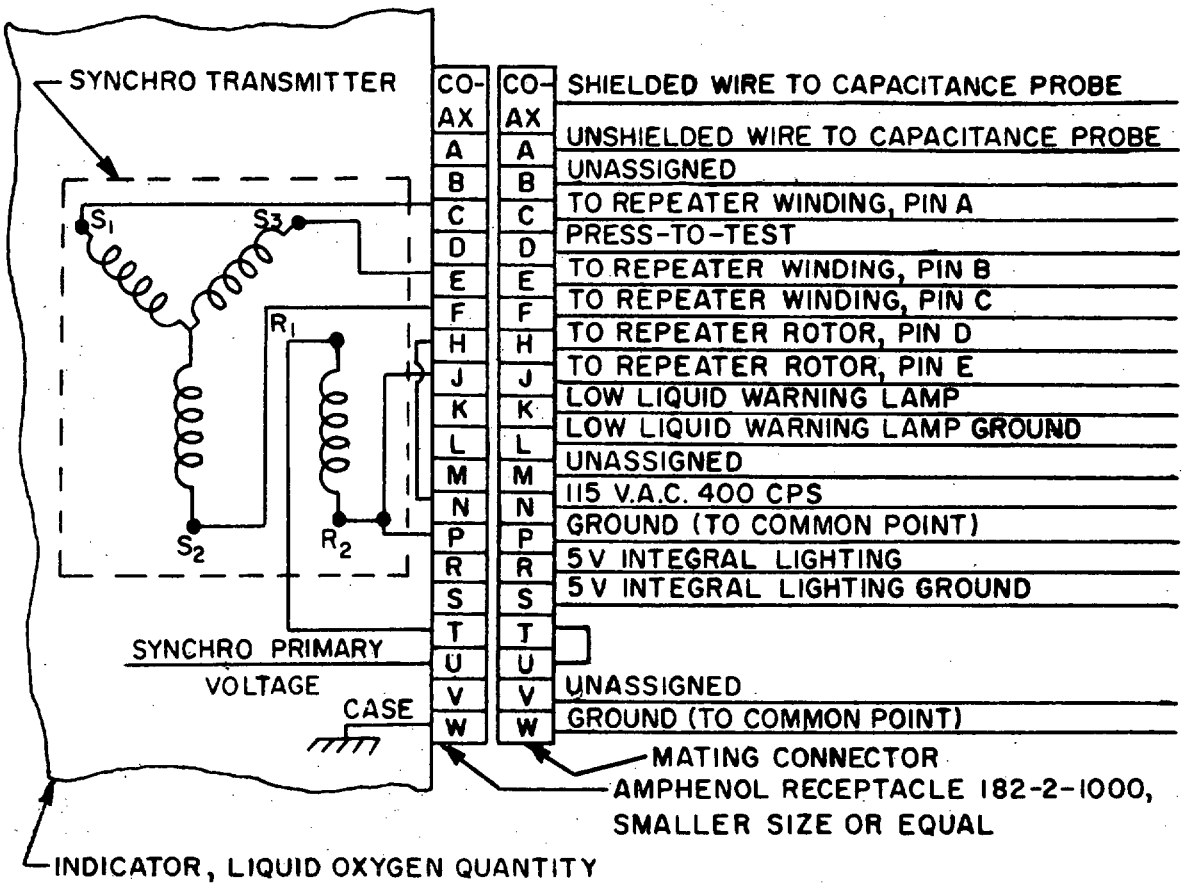
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CASE IN ACCORDANCE WITH MS33639, 2 INCH NOMINAL SIZE

FIGURE 3. INDICATOR CASE

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NOTES:
1. RECEPTACLE INSERT SHALL BE IN NORMAL POSITION.
2. ALTERNATE RECEPTACLES SHALL BE CAPABLE OF MATING WITH AMPHENOL PLUG 165-62, SMALL SIZE.
3. THE SYNCHRO TRANSMITTER SHALL BE WIRED TO THE RECEPTACLE AS SHOWN ABOVE.

FIGURE 4. ELECTRICAL RECEPTACLE

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3.6.12 Pointer - The pointer shall extend to, but not into the smallest graduations. The pointer shall be in accordance with MS 33585-8.

3.6.13 Power requirements - The indicator shall operate on 115 volt, single phase, 400 cycle AC that shall be obtained from a three-phase, four-wire "Y" connected, 115/200 volt grounded-neutral 400 cycle, AC electrical system, having the characteristics defined in MIL-STD-704, category B.

3.6.13.1 Synchro Transmitter - The synchro transmitter shall operate from a 26 or a 115 volt, single phase, 400 cycle AC power supplied to the primary. The power consumption of the transmitter shall not exceed 6.24 volt-amperes. The secondary voltage shall be 11.8 ± 0.3 volts and the variation between the three secondary voltages shall not exceed 0.2 volt.

3.6.14 Integral lighting - A 5 volt AC integral lighting system in accordance with MIL-L-25467 shall be provided.

3.6.15 Screw threads - Screw threads shall conform to MIL-S-7742.

3.7 Performance

3.7.1 Indicator scale error - The master test capacitor reading at each major dial graduation, when tested as specified in 4.7.2, shall be in accordance with Table I and Table II.

TABLE I INDICATOR QUANTITY CAPACITANCE

5 Liter Indicator		10 Liter Indicator		20 Liter Indicator		25 Liter Indicator	
Test Position (Liters)	*Capacitance (pf)	Test Position (Liters)	*Capacitance (pf)	Test Position (Liters)	*Capacitance (pf)	Test Position (Liters)	*Capacitance (pf)
0	63.50	0	123.50	0	247.00	0	303.50
1	69.30	2	135.09	4	270.18	5	332.48
2	75.09	4	146.68	8	293.36	10	361.46
3	80.89	6	158.28	12	316.56	15	390.44
4	86.68	8	169.87	16	339.74	20	419.42
5	92.48	10	181.46	20	362.92	25	448.40

*The tolerances specified in Table II shall be used where applicable.

NOTE: The room temperature readings as obtained by the requirements of the indicator scale error test shall be used in determining all subsequent scale errors.

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TABLE II

INDICATOR QUANTITY CAPACITANCE TOLERANCES
(pico - farads)

TEST CONDITIONS	INDICATORS			
	5 Liter	10 Liter	20 Liter	25 Liter
Room Temperature (no prior temperature conditioning)	± 0.4	± 0.7	± 1.4	± 1.8
Room Temperature (after temperature conditioning)	± 0.5	± 0.9	± 1.8	± 2.3
+ 160 ± 2°F	± 0.6	± 1.0	± 2.0	± 2.5
-65 ± 2°F	± 0.6	± 1.0	± 2.0	± 2.5

3.7.2 Leakage - The leakage in the hermetic seal of the indicator, when tested as specified in 4.7.3, shall not exceed 0.1 micron cubic foot per hour at a differential pressure of approximately one atmosphere.

3.7.3 Electrical zero - The pointer of the indicator and indicator repeater, when tested as specified in 4.7.4, shall move in the same direction and the indicator pointer shall be within 1.0 percent of full scale indication with the indicator repeater reading.

3.7.4 Dielectric strength - The leakage current between any unassigned pin (not wired to any circuit) of the electrical receptacle and the indicator case, when tested as specified in 4.7.5, shall not exceed 1.0 milliampere.

3.7.5 Insulation resistance - The insulation resistance between any pin (except pins R, S and W) of the electrical receptacle and the indicator case, when tested as specified in 4.7.6, shall be a minimum of 20 megohms.

3.7.6 Power failure - The indicator, when tested as specified in 4.7.7, shall actuate the power failure warning device of the indicator to indicate "OFF" until power is again applied. No part of the power failure flag shall appear in the dial cut out when the input power is supplied to the indicator.

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3.7.7 Press-to-test and low liquid level warning circuit - The press to test circuit of the indicator, when tested as specified in 4.7.8, shall drive the pointer of the indicator to zero or below. When the pointer reaches 10 percent of full scale indication, as indicated on the dial, the low liquid level warning circuit shall be activated and shall remain activated while the pointer is below 10 percent of full scale indication. With the pointer traveling up scale from zero, the low liquid level warning circuit shall be deactivated between 10 and 20 percent of full scale indication.

3.7.8 Dead spot - The capacitance range required to produce indicator pointer movement, when tested as specified in 4.7.9, shall not exceed 1.0 percent of full scale indication. Full scale indication shall be the difference between the capacitance at zero indication and the capacitance at full indication.

3.7.9 Power supply variation - The indicator, when tested as specified in 4.7.10, shall not exceed the rated power indication at the same scale position by more than 1.0 percent of full scale indication.

3.7.10 Speed of response - The indicator pointer, when tested as specified in 4.7.11, shall move from empty to full indication in a maximum time of 15 seconds and from full to empty in a maximum time of 15 seconds.

3.7.11 Pointer overshoot - The indicator pointer, when tested as specified in 4.7.12, shall not overshoot the final position more than one time and shall return to the final position within 2 seconds.

3.7.12 High and low temperature scale error - The indicator, when tested as specified in 4.7.13, shall show no evidence of fogging on the cover glass during the conditioning period and at the completion of the conditioning period the capacitance at each major dial graduation shall be accordance with Table I and the tolerances shown in Table II.

3.7.13 Vibration - The indicator, when tested as specified in 4.7.14, shall not have pointer oscillation or pointer variation exceeding ± 1/32 inch and there shall be no evidence of mechanical or material failure.

3.7.14 Shock test - The indicator, when tested as specified in 4.7.15, shall show no evidence of mechanical or material failure.

3.7.15 Acceleration test - The indicator, when tested as specified in 4.7.16, shall show no evidence of mechanical or material failure.

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3.7.16 Radio noise suppression test - The indicator, when tested as specified in 4.7.17, shall not exceed the radio noise suppression limits specified in MIL-I-6181.

3.7.17 Magnetic effect - The deflection of the compass at any of the specified positions, when the indicator is tested as specified in 4.7.18, shall not exceed 1 degree.

3.7.18 Cycling - The indicator, when tested as specified in 4.7.19, shall not exceed the limits specified for room temperature (after temperature conditioning) in Tables I and II.

3.7.19 Thermal shock - The indicator, when tested as specified in 4.7.20, shall show no evidence of moisture penetration or material failure.

3.7.20 High and low temperature exposure - The indicator, when tested as specified in 4.7.21, shall require a capacitance in accordance with Table I and the tolerances shown in Table II at each major dial graduation.

3.7.21 Salt spray test - The indicator, when tested as specified in 4.7.22, shall show no evidence of mechanical or material failure.

3.8 Interchangeability - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.9 Weight - The weight of the indicator shall not exceed 1.5 pounds.

3.10 Finish

3.10.1 Aluminum alloy parts - Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625. If such parts are inside the hermetically sealed area, they need not be anodized.

3.10.2 Steel parts - Non-corrosion resistant steel parts shall be cadmium plated where practicable in accordance with QQ-P-416, type II, class 2.

3.11 Identification of product - The indicators shall be marked for identification in accordance with MIL-STD-130, except that the Federal Stock Number shall be omitted from the nameplate.

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3.11.1 Nameplate - A nameplate shall be securely attached to each separately mounted unit and shall be legibly and permanently marked with the following information:

INDICATOR, LIQUID OXYGEN QUANTITY
Type
115 Volts, 400 Cycle, Single Phase
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Manufacturer's Name of Trade-mark
U.S.

3.12 Workmanship - The indicators shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could adversely affect safety, performance, reliability, or durability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the indicator shall be classified as follows:

(a) Qualification inspection - Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products (see 3.1, 4.3, 4.3.1 and 6.4).

(1) Retention of qualification - Retention of qualification consists of periodic verification to determine compliance of the qualified product with the requirements of this specification (see 3.1, 4.3, and 4.3.2).

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(b) First article inspection - First article inspection consist of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item conforms to the requirements of this specification (see 3.2 and 4.4 through 4.4.2).

(c) Qualify conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.3 Qualification inspection - Qualification inspection shall consist of all the examinations and tests of this specification.

4.3.1 Qualification samples - Qualification samples shall consist of two indicators for which qualification is required. Samples shall be forwarded to the Supply Officer, Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. The samples shall be plainly identified by securely attached durable tags marked with the following information:

Samples for qualification inspection
INDICATOR, LIQUID OXYGEN QUANTITY
Manufacturer's designation or number
Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-I-81387A(AS) and number under authorization (reference authorizing letter and number, see 6.4).

4.3.2 Retention - The retention of qualification shall consist of periodic verification to determine compliance of the qualified indicator with the requirements of this specification. The time and method of periodic verification shall be specified by the activity responsible for the Qualified Products List and shall be included in the Notice of Qualification letter.

4.4 First article inspection - First article inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Indicator scale error
Leakage
Electrical zero

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Dielectric strength
Insulation resistance
Power failure
Press-to-test and low liquid level warning circuit
Dead spot
Power supply variation
Speed of response
High and low temperature scale error

4.4.1 First article samples - Unless otherwise specified, as soon as practicable, after the award of the contract or order, the manufacturer shall submit two indicators. The samples shall be representative of the construction, workmanship, components, and materials to be used during production. When a manufacturer is in continuous production of these units from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or the waiving of first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The first article inspection samples shall be furnished to the Government as directed by the contracting officer (see 6.2).

4.4.1.1 Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the cognizant Government activity. The approved indicators will be returned to the manufacturer for use in monitoring production.

4.5 Quality conformance inspection - The sampling and inspection levels shall conform to MIL-STD-105. Quality conformance inspection shall consist of the following examinations and tests:

Visual examination
Dimensions
Indicator scale error
Electrical zero
Leakage
Dielectric strength
Insulation resistance
Power failure
Press to test and low liquid level warning circuit
Dead spot
Pointer overshoot
Preparation for delivery

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4.5.1 Sampling

4.5.1.1 Inspection lot

4.5.1.1.1 Indicator - An inspection lot size shall be expressed in units of indicators, made essentially under the same conditions and from the same materials and components. The sample unit shall be one indicator.

4.5.1.1.2 Preparation for delivery - An inspection lot size shall be expressed in units of one fully prepared shipping container, containing indicators fully prepared for delivery, made from essentially the same materials and components. The sample unit shall be one shipping container, containing indicators fully prepared for delivery with the exception that it need not be sealed.

4.5.1.2 Sampling for tests and examinations of indicators - The sample size, acceptance criteria, tests, and examinations required for the indicators shall be as specified in Table III.

4.6 Test conditions

4.6.1 Temperature and pressure - Unless otherwise specified, tests shall be conducted at local ambient temperature and barometric pressure. The temperature and barometric pressure shall be recorded at the time of inspection. This information shall be available for computation of test data, when required, to normal temperature and pressure (NTP) conditions. NTP conditions are 29.92 inches of mercury and 70°F. Test instruments shall be calibrated or adjusted according to their required usage in conducting individual tests.

4.6.2 Tapping - Unless otherwise specified, the indicator may be lightly tapped before a test reading is taken provided the test reading is within 2% of full scale indication to the desired indicator reading.

4.6.3 Test position - Unless otherwise specified, all tests shall be conducted with the indicator in the normal operating position.

4.6.4 Rated power - Unless otherwise specified, the indicator shall be tested with an 115 volt, 400 cycle, single-phase, AC power supply. The voltage and frequency tolerance shall be $\pm$ 2%.

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TABLE III

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS OF THE INDICATORS

INSPECTION	PARAGRAPH		SAMPLE SIZE	ACCEPTANCE CRITERIA 1/
	REQUIREMENT	METHOD		
Visual examination (see classification of defects)	3.6 through 3.6.14, 3.9 and 3.11	4.7.1.1.	Every indicator for critical defects. Inspection Level II for minor defects.	Reject all units with any critical defects. An acceptable quality level of 2.5 defects per hundred units for minor defects.
Dimensions	3.66 and Figures 1, 2 and 3	4.7.1.1.1	Inspection Level S-2	An acceptable quality level of 1.0 defect per 100 units.
Indicator Scale error	3.7.1	4.7.2	Every indicator	Reject all defective units.
Leakage	3.7.2	4.7.3	Every indicator	Reject all defective units.
Electrical Zero	3.7.3	4.7.4	Every indicator	Reject all defective units.
Dielectric strength	3.7.4	4.7.5	Every indicator	Reject all defective units.
Insulation resistance	3.7.5	4.7.6	Every indicator	Reject all defective units.
Power failure	3.7.6	4.7.7	Every indicator	Reject all defective units.

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TABLE III (Cont'd)

SAMPLE SIZE, ACCEPTANCE CRITERIA, TESTS AND EXAMINATIONS OF THE INDICATORS

INSPECTION	PARAGRAPH		SAMPLE SIZE	ACCEPTANCE CRITERIA 1/
	REQUIREMENT	METHOD		
Press to test and low liquid level warning circuit	3.7.7	4.7.8	Every indicator	Reject all defective units.
Dead spot	3.7.8	4.7.9	Every indicator	Reject all defective units.
Pointer overshoot	3.7.11	4.7.12	Inspection Level S-2	An acceptable quality level of 1.0 defect per 100 units.
Preparation for delivery	Section 5	4.7.1.2	Inspection Level S-2	An acceptable quality level of 2.5 defects per 100 units.

1/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

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4.6.5 Master test capacitor - The master test capacitor shall be a precision variable capacitor which shall have an accuracy of 0.2% or 0.2 pf, whichever is the greater, at a temperature of 25° C. In all tests and measurements specified herein, the tolerance of the master test capacitor shall be added to the specified tolerance for the given test or measurement.

4.6.6 Synchro electrical zero - With the synchro primary voltage (26 or 115 volts, 400 cps, single phase) applied to the synchro rotor leads, R_1 and R_2, and S_2 and R_2 connected to ground, the following conditions shall exist at electrical zero:

a. The voltage between the remaining stator terminals designated S_1 and S_3 shall be a minimum (usually 30 to 50 millivolts).

b. The voltage between R_1 and S_3 shall be near the low null (approximately 15.8 V for a 26 V synchro and 104.8 V for a 115 V synchro) and shall reach the low null (approximately 14.2 V for a 26 V synchro and 103.2 V for a 115 V synchro) when the rotor is rotated clockwise approximately 30° from electrical zero.

c. Clockwise rotation of the rotor not to exceed 30° from electrical zero shall cause an increase in the voltage between S_3 and S_2 and a decrease in the voltage between S_2 and S_1.

d. The voltage E (S_3S_2) shall be approximately in phase with the rotor voltage E(R_2R_1). The sequence of the terminal designations in parentheses indicates the direction of the voltage vector.

e. When the rotor is turned counterclockwise not to exceed 180° from electrical zero, the voltage E(S_1S_3) shall remain approximately in phase with the rotor voltage E(R_2R_1).

4.7 Inspection methods

4.7.1 Visual examination

4.7.1.1 Indicator - Every indicator shall be examined visually (for critical defects) to determine conformance to this specification. The classification of defects, Table IV, shall be used to classify the defects found.

4.7.1.1.1 Dimensions - The indicator shall be checked dimensionally to determine conformance to the dimensions specified herein and on the applicable Military Standard.

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TABLE IV

CLASSIFICATION OF DEFECTS FOR VISUAL EXAMINATION OF THE INDICATORS

Critical	Minor
1. Material imperfections-foreign matter embedded	201. Marking-missing, insufficient, incorrect, illegible, or not permanent.
2. Surface - unclean, rough, misaligned, or containing cracks, nicks, or other flaws.	202. Color not as specified.
3. Any component missing, malformed, fractured, or otherwise damaged.	
4. Any component loose or otherwise not securely retained.	
5. Incorrect assembling or improper positioning of components.	
6. Any functioning part that works with difficulty.	
7. Faulty workmanship or other irregularities.	

4.7.1.2 Preparation for delivery - Each of the fully prepared shipping containers, containing indicators selected as a sample unit from the lot shall be examined to determine that the packaging, packing, and marking conform to this specification. The classification of defects, Table V shall be used to classify the defects found.

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TABLE V

LIST OF DEFECTS FOR PREPARATION FOR DELIVERY

Item	Defects
Exterior and interior markings	Missing, incorrect, incomplete, illegible; of improper size, location, sequence; or method of application; markings not the same on the interior and exterior containers.
Packaging and Packing materials	Any non-conforming component; any component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of the components such as incomplete closure of the unit package, intermediate package, container flaps, loose strappings, etc.; bulging or distortion of the containers.
Exterior and interior weight of content	Number per container is more or less than required; gross or net weight exceeds the requirements.

4.7.2 Indicator scale error - The indicator shall be connected to rated power and the master test capacitor. The indicator pointer shall be set at the center of each major dial graduation (see Figures 1 and 2) by adjusting the capacitance of the master test capacitor. The capacitance required to set the pointer at the center of each major dial graduation shall be noted at each position and shall pass the requirements specified in 3.7.1

4.7.3 Leakage - A mass spectrometer type of helium leak detector shall be used as the method of measuring the leakage. The indicator shall pass the requirements specified in 3.7.2.

4.7.4 Electrical zero - The indicator shall be connected to rated power, master test capacitor and an indicator repeater. The indicator repeater shall meet the electrical characteristics of MIL-I-81388 and shall be subjected to and pass the electrical zero test of MIL-I-81388 prior to being connected to the indicator. By adjusting the capacitance of the master test capacitor, set the indicator pointer at the center of the zero indication and then to the full indication. The indicator shall pass the requirements specified in 3.7.3.

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4.7.5 Dielectric strength - 500 volts shall be applied between any unassigned pin (not wired to any circuit) of the electrical receptacle and the case of the indicator for a minimum of 5 seconds. The indicator shall pass the requirements specified in 3.7.4. If the indicator case is filled with an inert gas and hermetically sealed, the potential shall be reduced to 300 volts and the same test conducted.

4.7.6 Insulation resistance - The insulation resistance between any pin (except pins R, S and W) of the electrical receptacle that is wired to the indicator circuit and the case of the indicator shall be measured at 50 volts, direct current. The indicator shall pass the requirements specified in 3.7.5.

4.7.7 Power failure - The indicator shall be operated at rated power. The input power shall be interrupted for approximately 30 seconds and then the power shall be reapplied. The indicator shall pass the requirements specified in 3.7.6.

4.7.8 Press-to-test and low liquid level warning circuit - The indicator shall be connected to the master test capacitor and rated power. The low liquid level circuit shall be connected to suitable lights or other suitable devices. The master test capacitor shall be set to indicate a full reading on the indicator. The press-to-test circuit shall then be tested by grounding the press-to-test pin causing the pointer to move downward. When the pointer movement reaches zero or below, the ground shall be removed from the press-to-test pin. The indicator shall pass the requirements specified in 3.7.7.

4.7.9 Dead spot - The indicator shall be connected to the master test capacitor and rated power. The input capacitance to the indicator shall be slowly varied with the master test capacitor, first in one direction and then in the other direction. The capacitance change required to produce indicator pointer rotation shall be noted and shall be computed as a percentage of the full scale indication for which the indicator is calibrated. The indicator shall pass the requirements specified in 3.7.8.

4.7.10 Power supply variation - The indicator shall be connected to the master test capacitor and rated power. The following combinations of frequency and voltage of the rated input power shall be applied with the indicator pointer at approximately zero, midscale and full;

(a) 102 ± 1.0 volts 380 ± 5 cps
(b) 124 ± 1.0 volts 420 ± 5 cps

The indicator shall pass the requirements specified in 3.7.9. The indicator shall then be subjected to and pass the dead spot test except the frequency and voltage shall be as stated above and tested only at midscale.

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4.7.11 Speed of response - The indicator shall be connected to the master test capacitor and rated power. The master test capacitor shall be adjusted at room temperature so that the indicator pointer travels from zero to full indication and back to zero indication. The indicator shall pass the requirement specified in 3.7.10.

4.7.12 Pointer overshoot - The indicator shall be connected to the master test capacitor and rated power. The master test capacitor shall be adjusted to capacitances equal to each major dial graduation of the indicator. The indicator shall pass the requirements specified in 3.7.11.

4.7.13 High and low temperature scale error - The indicator shall be placed in a temperature chamber and subjected to a temperature of 160 ± 5°F for a minimum period of 4 hours. At the end of this period and while still at this temperature, input power shall be applied, the indicator pointer set at the center of each major dial graduation by means of adjusting the master test capacitor and recording the capacitance reading. The indicator shall then be returned to room temperature and the same scale readings repeated. The indicator shall pass the requirements specified in 3.7.12. The indicator shall then be subjected to and pass the following tests:

a. Press-to-test and
low liquid level warning circuit
b. Power failure

The above test procedure shall then be repeated utilizing a temperature of -65 ± 5°F for a minimum period of 4 hours.

4.7.14 Vibration - The indicator shall be subjected to the vibration test specified in MIL-E-5272, Procedures IV, and V. Procedure IV shall be conducted with the indicator energized and the pointer set at the following positions: approximately zero point, midscale, and the upperscale end. The indicator shall pass the requirements specified in 3.7.13. The indicator shall then be subjected to and pass the indicator scale error test.

4.7.15 Shock test - The indicator shall be subjected to the shock test (equipment operation) specified in MIL-E-5272, Procedure V. The indicator shall pass the requirements specified in 3.7.14. The indicator shall then be subjected to and pass the following tests:

a. Indicator scale error
b. Press-to-test and
low liquid level warning circuit
c. Power failure
d. Sealing

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4.7.16 Acceleration test - The indicator shall be subjected to the acceleration test specified in MIL-E-5272, Procedure III. The applied acceleration shall be 11G. The indicator shall pass the requirements specified in 3.7.15. The indicator shall then be subjected to and pass the following tests:

a. Indicator scale error
b. Press-to-test and low liquid level warning circuit
c. Power failure
d. Sealing

4.7.17 Radio noise suppression test - The indicator shall be subjected to the radio noise suppression test as specified in MIL-I-6181. The indicator shall pass the requirements specified in 3.7.16.

4.7.18 Magnetic effect test - The indicator, not operating, shall be rotated in a vertical plane about a short bar magnet compass with the nearest part of the indicator 5-1/2 inches from, and magnetically east or west of, the center of the compass. Starting directly under compass, the indicator shall be held in positions 0, 45, 90, 135, 180, 225, 270, and 315 degrees from the initial position. At each of these positions the indicator shall be rotated on its own horizontal axis until it is in its normal upright position. The horizontal magnetic field intensity shall be 0.17 to 0.19 gauss. The indicator shall pass the requirements specified in 3.7.17. The indicator shall then be subjected to the same test with input power supplied.

4.7.19 Cycling - The indicator shall be connected to the master test capacitor and then subject to a minimum of 5000 cycles. Each cycle shall consist of continuously varying the capacitance so that the pointer moves from zero to full and return to zero within 5 ± 1 minutes. After each 12 cycles, the indicator shall be de-energized electrically for a minimum of 10 minutes. At the conclusion of the cycling, the indicator shall be subjected to the indicator scale error test. The indicator shall pass the requirements specified in 3.7.18.

4.7.20 Thermal shock - The indicator shall be subjected to eight cycles of immersion in tap water maintained at 185 ± 2°F and 40 ± 2°F. Each cycle shall consist of a 30 minute immersion in water maintained at 185 ± 2°F followed by a 30 minute immersion in water maintained at 40 ± 2°F and not more than 5 seconds shall elapse between immersions. The indicator shall pass the requirements specified in 3.7.19. The indicator shall then be subjected to and pass the sealing test.

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4.7.21 High and low temperature exposure - The indicator shall be subjected to a temperature of 160 ± 5°F for a minimum period of 24 hours. At the end of this period, the indicator shall be returned to room temperature, input power applied, and the indicator pointer set at the center of each major dial graduation by adjusting the master test capacitor. The readings on the master test capacitor shall be noted. The indicator shall pass the requirements specified in 3.7.20. The indicator shall then be subjected to and pass the following tests:

a. Press-to-test and low liquid level warning circuit
b. Power failure

The above test procedure shall then be repeated utilizing a temperature of -65 ± 5°F for a minimum period of 24 hours.

4.7.22 Salt spray test - The indicator shall be subjected to the salt spray test specified in MIL-E-5272, Procedure I. The indicator shall pass the requirements specified in 3.7.21.

5. PREPARATION FOR DELIVERY

5.1 Packaging - The indicator shall be packaged in accordance with MIL-STD-794, Level A or C as specified in contract or order (see 6.2). For Level A packaging the method of preservation shall be in accordance with MIL-P-116, Method 1A-5, without preservative compound (using metal reusable containers).

5.2 Packing - The indicator shall be packed in accordance with MIL-STD-794, Level A, B, or C as specified in contract or order (see 6.2).

5.3 Marking - The interior and exterior containers shall be marked in accordance with MIL-STD-794.

5.3.1 Interior container - Each interior container shall be marked with the following information:

INDICATORS, LIQUID OXYGEN QUANTITY
Specification MIL-I-81387
Manufacturer's Part No.
Manufacturer's Serial No.
Contract or Order No.
Date of Manufacture
Name of Manufacturer

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Name of Contractor (if different from Manufacturer)
"IF IN STORAGE AFTER (DATE)*, THE INDICATOR SHALL BE TESTED AND INSPECTED BEFORE USE."

NOTE: *This date shall be 1 year from the date of inspection and shall be inserted by the Inspector at the time of acceptance.

5.3.2 Precautionary marking - The following precautionary markings shall appear conspicuously on two opposite sides of each interior package and shipping container wherever practicable.

FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE

6. NOTES

6.1 Intended use - The liquid oxygen quantity indicator covered by this specification is intended to indicate the quantity of liquid oxygen contained in an aircraft liquid oxygen system.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of indicator required (see 1.2).
(c) Whether first article inspection is required and where the sample should be delivered (see 4.4.1).
(d) Selection of applicable levels of preservation, packaging, and packing.
(e) Applicable methods of cleaning and preservation.
(f) Items of Data required (see 6.3).

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVAIR Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

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6.4 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Commander, Naval Air Systems Command, Department of the Navy, Washington, D. C. 20360; however, authorization for qualification of products shall be obtained from the Commanding Officer, Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Director, Aerospace Crew Equipment Department. Prior to submission of the samples for qualification inspection, the manufacturer shall submit a request to the Aerospace Crew Equipment Department indicating a date on which the samples can be forwarded and also request an authorization number to accompany the samples.

6.4.1 When requested, the manufacturer shall submit engineering drawings and inspection reports in accordance with SD-6, Provisions Governing Qualification.

6.5 International standardization - Certain provisions of this specification are the subject of international standardization agreement (ABC 10/2). When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Officers so that appropriate action may be taken respecting the international agreement concerned.

Project No. 1660-N030

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-I-81387A(AS) INDICATORS, LIQUID OXYGEN QUANTITY

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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